

A

Mini Project Report (BCC 351) (3rd Semester)

On

“DIGITAL QUEUE MANAGEMENT SYSTEM”

Group No-25

Submitted

In partial fulfillment

For the award of the Degree of

Bachelor of Technology

In the Department of Computer Science & Engineering



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Department Of Computer Science & Engineering

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November 2025

Vision and Mission of the Institute and Department

Vision:

To make IMSEC an Institution of Excellence for empowering students through technical education, incorporating human values, and developing engineering acumen for innovations and leadership skills to upgrade society.

Mission:

Mission 1: To promote academic excellence by continuous learning in core and emerging Engineering domains using innovative teaching and learning methodologies.

Mission 2: To inculcate values and ethics among the learners.

Mission 3: To promote industry interactions and cultivate young minds for entrepreneurship.

Mission 4: To create a conducive learning ecosystem and research environment on a perpetual basis to develop students as technology leaders and entrepreneurs who can address tomorrow's societal needs.

Vision of the Department

To provide globally competent professionals in the field of Computer Science & Engineering embedded with sound technical knowledge, aptitude for research and innovation, and nurture future leaders with ethical values to cater to the industrial & societal needs.

Mission of the Department

Mission 1: To provide quality education in both the theoretical and applied foundations of Computer Science & Engineering.

Mission 2: Conduct research in Computer Science & Engineering resulting in innovations thereby nurturing entrepreneurial thinking.

Mission 3: To inculcate team building skills and promote life-long learning with a high societal and ethical values.

Program Outcomes (POs) and Program Specific Outcomes (PSOs)

S. No.	Program Outcomes / Program Specific Outcomes
PO1.	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2.	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3.	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4.	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5.	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
PO6.	The engineer and society: apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7.	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8.	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9.	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10.	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11.	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12.	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

CSE Department Program Educational Objectives (PEOs)

B. Tech Computer Science & Engineering Department has the following Program Educational Objectives:

PEO1: Possess core theoretical and practical knowledge in Computer Science and Engineering for successful career development in industry, pursuing higher studies or entrepreneurship.

PEO2: Ability to imbibe lifelong learning for global challenges to impact society and environment.

PEO3: To demonstrate work productivity with leadership and managerial skills having ethics and human value in progressive career path.

PEO4: To exhibit communication skill and collaborative skill plans and participate in multidisciplinary Computer Science & Engineering fields. .

CSE Department Program Specific Outcomes (PSOs)

B. Tech Computer Science & Engineering Department has the following Program Specific Outcomes:

PSO1: To analyze and demonstrate, the recent engineering practices, ethical values and strategies in real time world problems to meet the challenges for the future.

PSO2: To develop adaptive computing system using computational intelligence strategies and algorithmic design to address diverse data analysis and machine learning challenges.

CO-PO-PSO MAPPING FOR ACADEMIC SESSION 2025-26

Course Name: Mini Project

Course Code: BCC-351

Semester / Year: 3rd / 2

NBA Code: C209

Faculty name(s): Ms. Rinki Singh / Ms. Swati Singh

Course Outcomes:

SL.NO	DESCRIPTION	COGNITIVE LEVEL (BLOOMS TAXONOMY)
C209.1	Developing a technical artifact requiring new technical skills and effectively utilizing a new software tool to complete a task	K3
C209.2	Writing requirements documentation, Selecting appropriate technologies, identifying and creating appropriate test cases for systems.	K4
C209.3	Demonstrating understanding of professional customs & practices and working with professional standards.	K2
C209.4	Improving problem-solving, critical thinking skills and report writing.	K5
C209.5	Learning professional skills like exercising leadership, behaving professionally, behaving ethically, listening effectively, participating as a member of a team, developing appropriate workplace attitudes	K6

CO-PO-PSO Mapping:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
C209.1	3	3	3	3	3	2	1	2	2	2	2	3	3	3
C209.2	3	3	2	2	2	1	1	2	2	3	1	1	2	3
C209.3	1	1	1	1	1	3	2	3	2	2	2	2	2	1
C209.4	3	3	3	3	3	2	1	1	2	3	1	3	1	1
C209.5	1	1	1	1	1	2	2	3	3	3	3	1	1	1
C209	2.20	2.20	2.00	2.00	2.00	2.00	1.40	2.20	2.20	2.60	1.80	2.00	1.80	1.80

Candidate's Declaration

I hereby declare that the work, which is being presented in the Mini Project (BCC 351), titled “**DIGITAL QUEUE MANGEMENT SYSTEM**” is being submitted for the partial fulfillment for the Mini project work in 3rd semester for the award of “**Bachelor of Technology**” in the **Department of Computer Science & Engineering, IMS Engineering College, Ghaziabad**, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh. This mini project work is a record of my own investigations carried under the Guidance of **Mr. Nishant Anand, Assistant Professor**, IMS Engineering College, Ghaziabad.

I have not submitted the project work presented in this Mini project anywhere for the award of any other Degree.

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CERTIFICATE

This is certified that the Mini Project Report titled “**Digital Queue Management System**” submitted by **Sparsh Dixit, Shiva Sharma, Suryansh Mishra, Rahul Singh, Md. Anas** in the **Department of Computer Science and Engineering, IMS Engineering College, Ghaziabad**, affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow, Uttar Pradesh for the award of the Degree of Bachelor of Technology in Computer Science & Engineering is a record of Bonafide work carried out by him/her/them under my supervision. In my opinion, the 3rd Semester Mini Project Report has reached the standards of fulfilling the requirements of the regulations of Degree.

Mr. Nishant Anand

Assistant Professor

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IMS Engineering College, Ghaziabad

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my Mini Project Supervisor **Mr. Nishant Anand, Assistant Professor**, IMS Engineering College, Ghaziabad, who gave me this opportunity with important guidelines regarding various points to be taken for my Mini Project work and also for his deep involvement, precious advice, invaluable suggestions and continuous motivation throughout this work. I am highly obliged to him for always being there for my needs. I consider myself very fortunate for having been associated with a Mentor like him. His affection, guidance, and scientific approach served as a veritable incentive for the completion of this work.

I am extremely thankful to **Prof. (Dr.) Manas Kumar Mishra, Director, IMSEC**, Ghaziabad, for providing me with infrastructural facilities to work in, without which this work would not have been possible.

I express my sincere gratitude to **Prof. (Dr.) Sonali Mathur, HoD(CSE)**, IMSEC, Ghaziabad, for his stimulating guidance, continuous encouragement, and supervision throughout the course of the present work.

I would also like to express my genuine gratitude to **Ms. Rinki Singh and Ms. Swati Singh, Mini Project Coordinator** for their valuable suggestions and advice in carrying out this work.

I also like to thank the entire department faculty who helped me directly or indirectly to complete my work. This acknowledgment will remain incomplete if I fail to express my deep sense of obligation to my family and God for their consistent blessings and encouragement.

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CHAPTER 1

INTRODUCTION

1.1 Background

Queue management has become one of the essential focus areas in modern service-driven sectors such as hospitals, banks, government offices, retail stores, and educational institutions. In traditional setups, people are required to wait in long physical lines, which leads to frustration, time wastage, and inefficiency. These long queues often result in overcrowding, miscommunication, service delays, and discomfort for customers, especially the elderly, patients, and busy professionals.

Despite rapid digitalization, many organizations still rely on manual queue systems such as paper tokens or physical waiting lines. These outdated methods create administrative burdens and reduce customer satisfaction. Moreover, after the COVID-19 pandemic, maintaining social distancing and minimizing physical contact have become crucial, highlighting the need for smarter and more hygienic queue management solutions.

In recent years, the demand for systematic, contactless, and real-time queue management has increased rapidly across various sectors. Customers today expect transparency, convenience, and shorter waiting times. However, most traditional organizations lack the digital infrastructure to provide real-time queue updates, remote booking, or automated service allocation. This creates a significant gap between customer expectations and actual service delivery.

A Digital Queue Management System (DQMS) is designed to bridge this gap by providing a smart, technology-driven platform that organizes queues efficiently and enhances the overall customer experience. It enables users to join queues virtually, monitor waiting time through mobile apps or display screens, receive notifications, and reach the service counter only when their turn approaches. This significantly reduces crowding, waiting-time anxiety, and operational inefficiency.

The core background of the system lies in the increasing need for automation across service industries. With advancements in IoT, mobile applications, and cloud computing, organizations can now manage high customer volumes without physical chaos. Traditional token systems or manual registrations are often inaccurate, slow, and prone to human error. A digital queue solution provides scalability, accuracy, and real-time analytics to improve workflow and optimize service delivery.

The proposed **Digital Queue Management System** addresses these challenges by introducing a centralized platform where:

- Customers can take e-tokens and join the queue from anywhere.
- Real-time updates about queue position and waiting time are provided.
- Service providers can manage, prioritize, and track queues efficiently.
- Crowd density is reduced, ensuring comfort and safety.
- Organizations gain insights through automated data reports and analytics.

By integrating modern technology, DQMS enhances service speed, reduces operational load, improves customer satisfaction, and supports digital transformation in both public and private sectors.

1.2 Objective

The primary objective of the Digital Queue Management System (DQMS) is to streamline and modernize the queue-handling process by providing customers a convenient, time-saving, and contactless way to manage waiting lines. The system aims to eliminate the need for physical queues and reduce overcrowding by allowing users to generate digital tokens, track their position in real time, and receive notifications when their turn is approaching.

The system promotes operational efficiency by automating token generation, service allocation, and queue monitoring. It also enhances customer satisfaction by minimizing waiting time, improving transparency, and enabling users to plan their visit better. Additionally, the project supports digital literacy by offering a simple interface where users can book slots, check updates, reschedule visits, and interact with service authorities easily. Ensuring accessibility for all users—whether students, patients, elderly individuals, or working professionals—is a key objective of this system.

1.3 Scope of our project

The system can be implemented in hospitals (for OPD and lab queues), government offices (for documentation and service counters), banks, retail stores, educational institutions, clinics, and corporate service desks. Customers can generate e-tokens, book appointments, track queue status, and receive real-time alerts. Service providers can view customer flow, manage multiple queues simultaneously, monitor staff performance, and handle crowded hours more effectively.

By reducing long physical lines, the project enhances comfort, safety, and service quality. The system contributes to faster operations, reduced waiting times, better crowd control, and improved user experience—making it a practical digital solution for modern service environments.

1.4 Problem Statement

Many service organizations, such as hospitals, banks, government offices, retail centers, and educational institutions, still rely on traditional manual queue systems that require customers to wait in long physical lines. This leads to frustration, time wastage, overcrowding, and inefficient service delivery. Due to the absence of a digital and organized platform, customers cannot track their waiting time, receive real-time updates, or plan their visit effectively. This results in poor user experience, miscommunication, and unnecessary congestion.

CHAPTER 2

LITERATURE SURVEY

- According to K. Rajesh and M. Devi (2022), providing web-based or mobile-based token generation allows patients to register without standing in queues. The authors emphasized that digital alerts to patients (SMS/notifications) help reduce physical presence inside hospitals and improve patient staff. Flow.
our system also implements real-time queue status visibility, making the process comfortable for patients
- A study by Abdullah & Saleh (2023) found that real-time queue analytics helps hospitals in resource management, such as distributing patients across multiple doctors and handling emergencies more efficiently.
In our project, admin can prioritize emergency cases and skip non-available patients improving service quality.
- WHO (World Health Organization) in their healthcare digitization report mentioned that automation in hospitals helps in maintaining hygiene and safety, especially during infections like COVID-19, where crowd control becomes essential
Our system reduces close physical interactions by enabling controlled queue calling.
- According to K. Rajesh and M. Devi (2022), providing web-based or mobile-based token generation allows patients to register without standing in queues. The authors emphasized that digital alerts to patients (SMS/notifications) help reduce physical presence inside hospitals and improve patient flow.
Our system also implements real-time queue status visibility, making the process comfortable for patients and staff.

CHAPTER 3

METHODOLOGY

3.1 Problem Definition:

Traditional queue systems in hospitals, banks, government offices and service centers often result in long waiting lines, overcrowding and poor customer experience. People spend a significant amount of time physically standing in queues without any idea of when their turn will come. This not only causes frustration and stress, but also affects productivity and service efficiency.

The primary issue is the absence of a smart system that can **digitally generate tokens, track queue status in real-time and notify users about their turn** without requiring physical presence. Lack of digitization makes customers wait unnecessarily, leading to time wastage and mismanagement.

Additionally, due to the manual queue handling system, staff members face difficulty in organizing visitors, prioritizing urgent cases, and maintaining records. Miscommunication, token duplication and delayed service further reduce service quality and user satisfaction.

Another major challenge is accessibility. Many users especially elderly people and those living far from the service location do not have access to platforms that provide estimated waiting time or online token booking. This results in overcrowding at peak hours and increases the workload of employees.

The **Digital Queue Management System Project** aims to solve these challenges by developing a user-friendly web platform that allows users to:

- generate tokens online
- track their token position in real-time
- receive notifications when their turn is near.

This system ensures **reduced waiting time, improved service delivery and enhanced user convenience** leading to better management and streamlined operations.

3.2 Data Flow Diagram:

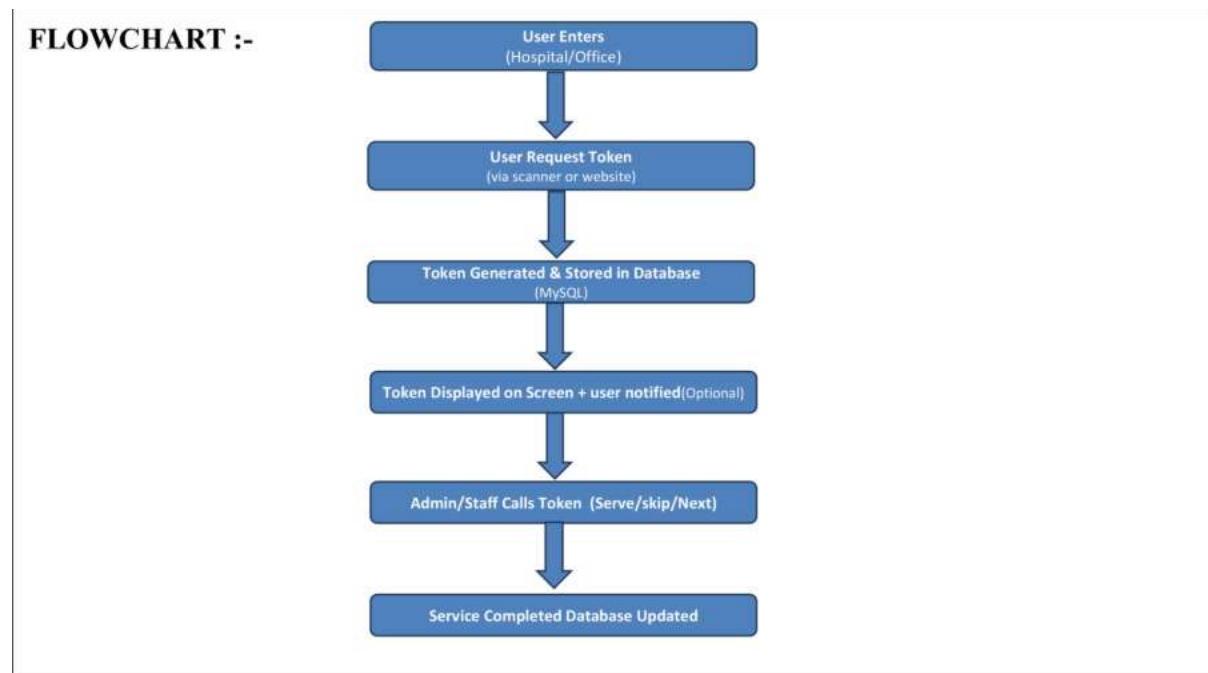


Figure 3.2 Data Flow diagram

Figure 3.2 show the Data Flow Chart

1. The process starts when a user arrives at the service location.
2. The user can request a token either by scanning a QR code / kiosk / website.
3. The system creates a unique token number and stores user + token details securely in the MySQL database.
4. The token number is shown instantly on the user's device/screen, and optionally a notification/update is sent.
5. The staff can call the next token for service using the admin dashboard. They also have the option to serve, skip, or recall tokens.
6. Once the service is completed, the token status is marked as Served and the database automatically updates queue status.

3.3 Tools and Technology Used:

a) HTML 5 (Hypertext Markup Language)

HTML5 is the language that builds the structure of webpages. It organizes content into sections, headers, forms, images, buttons, and links.

How we used it:

In our project, HTML5 was used to create the frame of every webpage — including the home page, product listing, product detail pages, user registration, login forms, and cart page.

- We structured pages using semantic tags: `<header>`, `<nav>`, `<main>`, `<section>`, `<footer>`.
- Forms like login and product upload are built with HTML5 inputs that support user data validation.

b) CSS3 (Cascading Style Sheets)

CSS3 styles the HTML structure — colours, fonts, layouts, spacing, and responsiveness.

How we used it:

CSS3 made the website visually attractive and user-friendly across devices.

- We used CSS Grid and Flexbox for arranging product cards side-by-side in a responsive grid.
- Colours and fonts were consistently applied using CSS variables for easy theming.
- Hover effects, button styles, and responsive breakpoints created smooth interactions and layouts for mobiles, tablets, and desktops.

c) JavaScript

JavaScript adds interactivity to the website. It updates queue numbers, validates forms, refreshes token status, and handles dynamic UI changes without reloading the page.

Used for: Real-time display updates + form validation + dynamic content loading

d) Flask (Python Framework)

Flask is used for the backend and server-side programming of the system.

It handles all business logic such as:

- Routing (Navigation between pages)

- Server communication and APIs
- Token generation and auto-fetching
- Authentication for admin & doctor logins

Flask also manages secure interaction between frontend and database using HTTP requests.

Used for: REST APIs, Login System, Queue Management Logic Implementation

e) **MySQL Database**

MySQL stores all hospital data in a structured and relational format.

Major tables include:

tokens (Token number, time, status)

patients (Patient details)

doctors (Doctor info & specialization)

users (Login credentials & roles)

Queries are used to perform CRUD operations, status updates, and retrieve real-time token information.

Used for: Patient records, queue tracking, admin monitoring, history logs

CHAPTER 4

IMPLEMENTATION

4.1 Website Preview

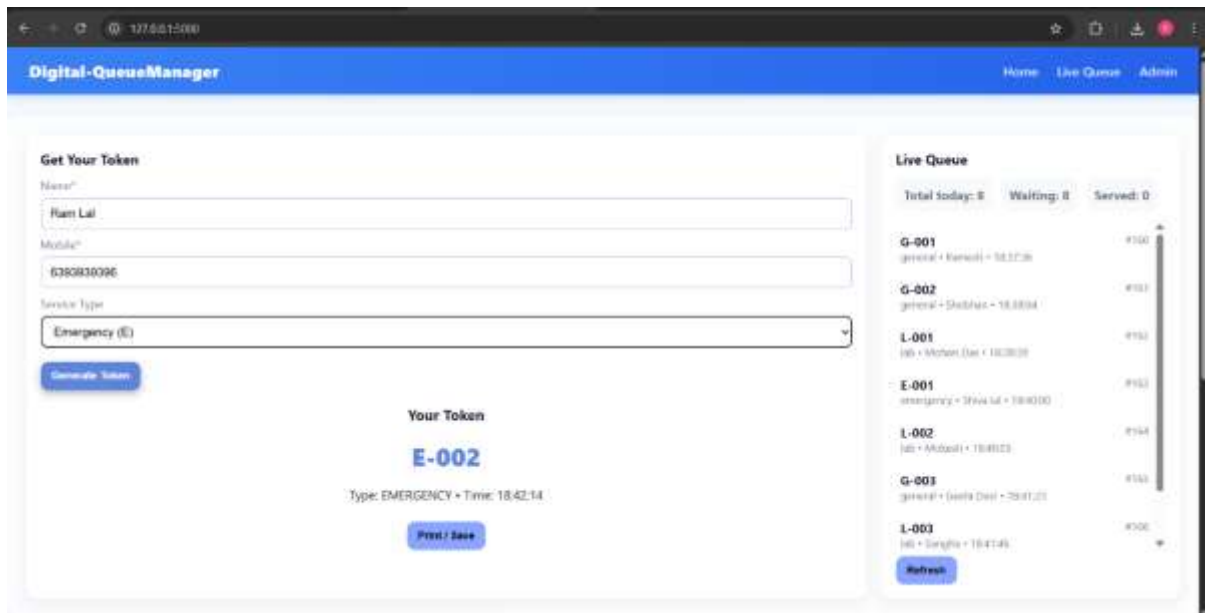


Figure 4.1: Home Page

Figure 4.1. shows the homepage of the “Digital Queue Manager for Hospital” system which is designed with a clean and user-friendly interface for quick hospital queue access. The layout allows patients to easily select the department or doctor and proceed to generate their token. The homepage also includes options like View Queue Status and Hospital Information, helping patients navigate the system smoothly without any confusion. The interface is lightweight, responsive, and accessible for users of all age groups, ensuring minimum waiting time and less crowding at the inquiry counter.

4.2 Admin Login Page :

Figures 4.2 shows the Admin Login interface of the Digital Queue Manager for Hospital system. This page is specifically designed for hospital staff such as the receptionist or queue manager. The administrator must enter valid login credentials like username and password to access the secure backend system.

The page is built with a clean and simple UI so that hospital staff can quickly enter the system without any technical difficulty. Proper validation and security are applied to avoid unauthorized access and protect sensitive patient data.

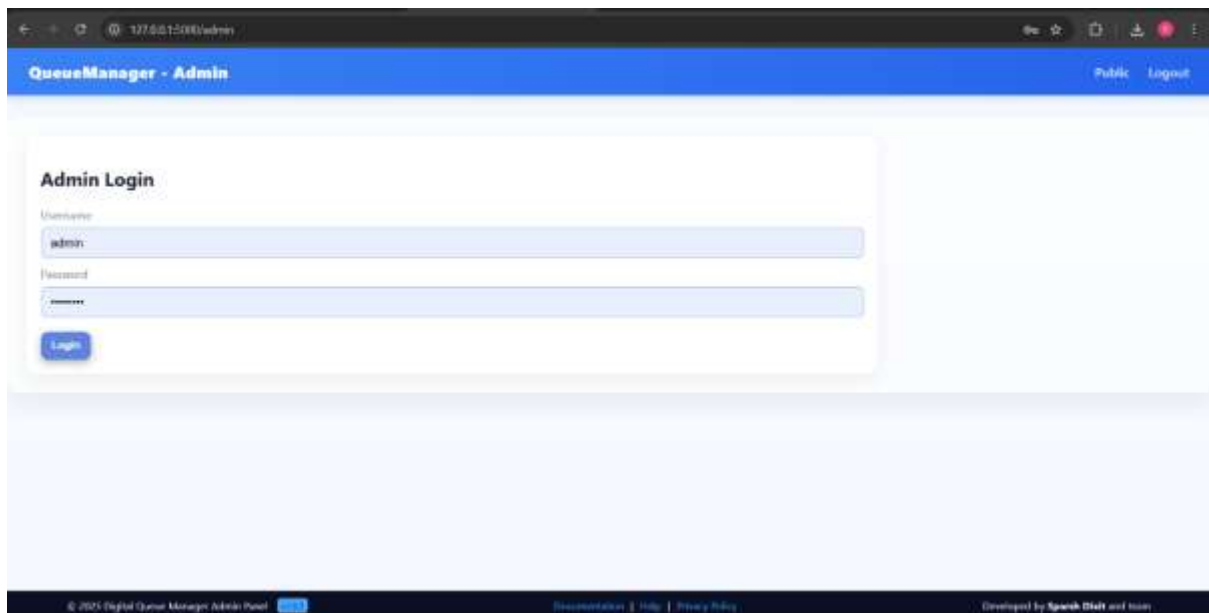


Figure 4.2: Admin Login Page

4.3 Token Serve Page:

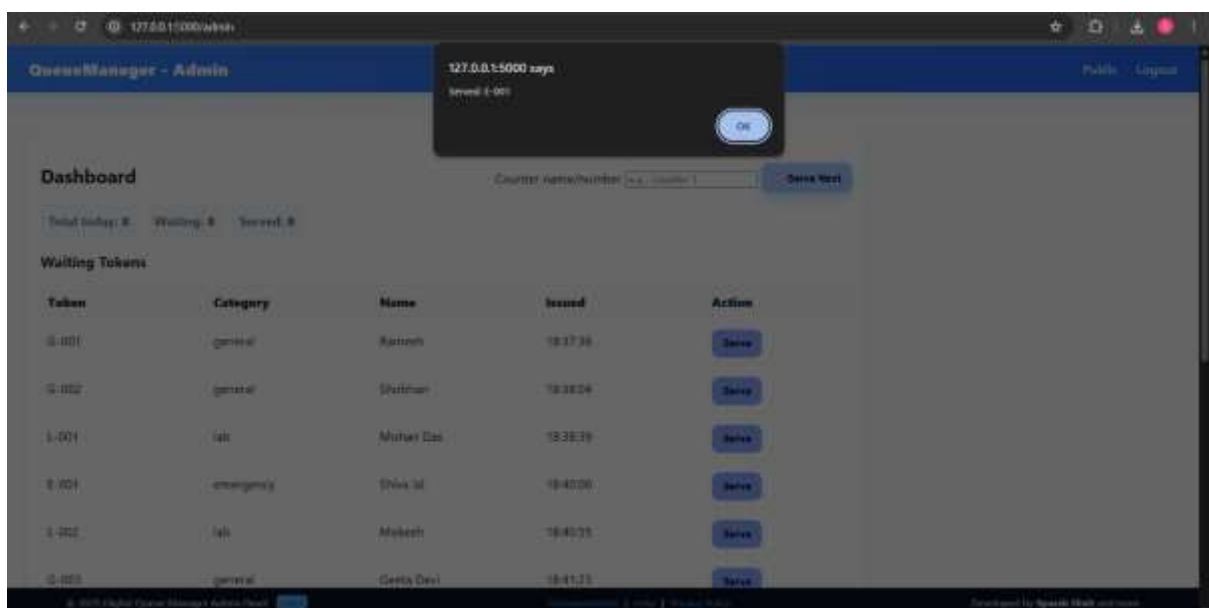


Figure 4.3: Token Serve Page

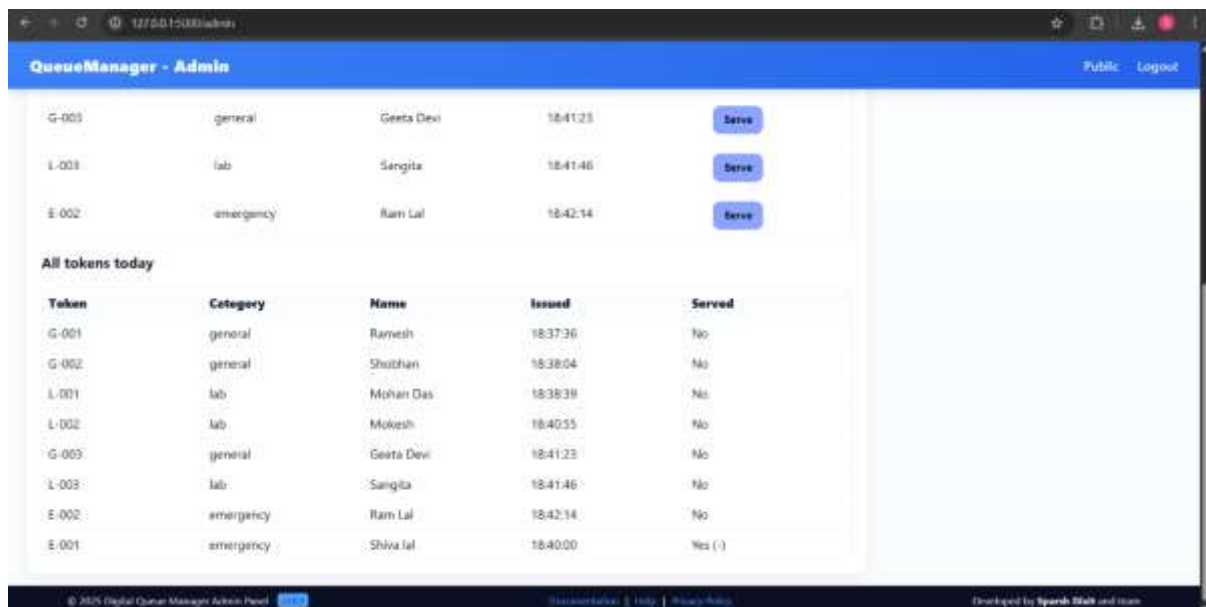
Figure 4.3 shows the Token Serving page of the Digital Queue Manager for Hospital system. This module is used by the hospital receptionist or admin to manage the flow of patients in real time. The admin can view the list of all allocated tokens along with their status such as Pending, Called, Skipped, or Completed.

When the admin clicks on the “Serve Token” button, the system automatically sends the next token number to the Digital Display Screen and Doctor Dashboard.

Additionally, the admin can:

- ✓ Skip a patient if not present
- ✓ Put a token on hold
- ✓ Mark the patient as served after consultation
- ✓ View queue length and waiting patients

4.4 View All Today Token Page :



The screenshot shows the 'QueueManager - Admin' interface. At the top, there's a blue header with 'QueueManager - Admin' and 'Public Logout'. Below the header, there's a list of tokens with columns for Token, Category, Name, Issued, and Served. The tokens are: G-001 (general, Geeta Devi, 18:41:23), L-001 (lab, Sangita, 18:41:46), and E-002 (emergency, Ram Lal, 18:42:14). Each token has a 'Serve' button. Below this list, there's a section titled 'All tokens today' which contains a table with the same columns as the list above. The table lists all tokens for the day, including G-001, G-002, L-001, L-002, G-003, L-003, E-002, and E-001. The 'Served' column shows 'No' for most tokens and 'Yes (-)' for E-001. At the bottom of the interface, there's a footer with '© 2025 Digital Queue Manager Admin Panel', 'Documentation | Help | Privacy Policy', and 'Developed by Sparsh Shah and team'.

Token	Category	Name	Issued	Served
G-001	general	Ramesh	18:37:36	No
G-002	general	Shobhan	18:38:04	No
L-001	lab	Mohan Das	18:38:39	No
L-002	lab	Mokesh	18:40:55	No
G-003	general	Geeta Devi	18:41:23	No
L-003	lab	Sangita	18:41:46	No
E-002	emergency	Ram Lal	18:42:14	No
E-001	emergency	Shiva Lal	18:40:00	Yes (-)

Figure 4.4: View all Today Token Page

Figure 4.4 shows the “View All Today’s Tokens” interface which is designed for hospital staff to monitor the complete list of patients registered for consultation on the current day. This module displays all token entries in a tabular format including details like token number, patient name, department, and current status (Yes/No).

This page helps the admin and reception team keep track of:

Total patient load for the day

Tokens served and remaining

Current queue progress

4.5 Footer Page:

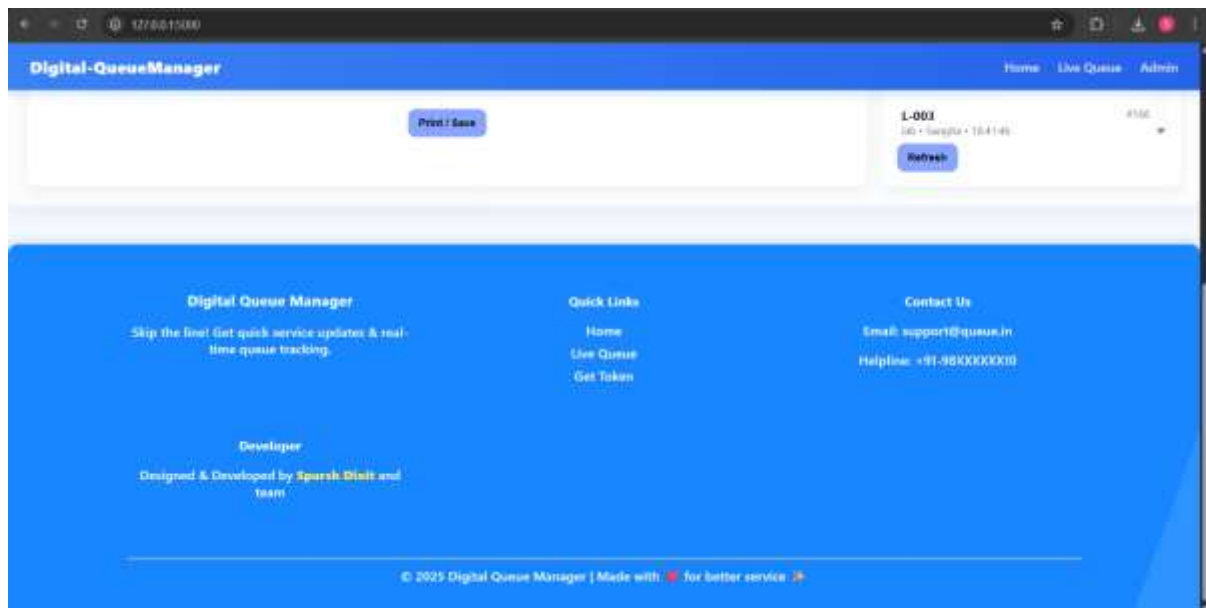


Figure 4.5: Footer Page

Figure 4.5 shows the footer section of the Digital Queue Manager for Hospital system. The footer is designed to provide helpful links, contact information, and credits to the developer in a clean and organized layout. It remains visible at the bottom of every page, ensuring easy navigation and quick access to essential information.

The footer contains the project name, a brief tagline, and useful quick links such as Home, Live Queue Status, and Get Token options for patients. It also highlights the developer credit with a personalized note, maintaining professionalism and ownership of the project work.

Additionally, the footer provides a helpline number and support email address so that users can reach out if they face any issues while generating tokens or navigating the portal.

The overall design uses a hospital-theme color scheme, maintaining readability and modern UI standards. A copyright notice is also included to indicate the system's development year and intellectual ownership.

CHAPTER 5

FUTURE SCOPE & CONCLUSION

5.1 LIMITATION OF THE SYSTEM

1. The system works properly only when internet or LAN connection is stable; offline support is limited.
2. Integration with real-time hospital hardware tools like digital display boards, token kiosks, and calling systems is not fully implemented.
3. The system currently uses basic authentication and does not include advanced security practices such as multi-factor login and encrypted data handling.
4. Communication features such as SMS or WhatsApp notifications for token updates are not yet added.
5. The performance has not been tested at large-scale hospitals with multiple OPD counters and heavy patient loads.
6. The system supports limited user roles; more roles such as Nurse, Reception Supervisor, and Emergency Desk can be introduced.

5.2 FUTURE SCOPE

1. Smart reporting and analytics dashboard to show patient load, doctor schedule, and waiting time predictions.
Automatic token calling integration with speakers and digital screens.
2. Mobile app for patients to check live queue status from home.
3. SMS / WhatsApp alerts for token number, expected waiting time, and doctor availability.
4. Predictive analysis for reducing patient crowd in peak hours
Auto-distribution of patients to doctors with minimum waiting time.
5. Single app to manage queue across multiple hospitals / clinics.
6. Online bill payment and OPD slip generation. Integration with E-Hospital/ABHA health record system.

5.3 CONCLUSION

The Digital Queue Manager for Hospital successfully addresses one of the major challenges in hospitals — long waiting lines and unorganized appointment flow. By generating digital tokens, showing real-time queue status, and improving communication between patients and hospital staff, the system reduces rush and panic in hospitals.

This solution enhances the overall patient experience, ensures faster service, and helps doctors manage appointments more efficiently. Digital queueing also minimizes physical crowding, contributing to improved hygiene and safety — especially important in cases like COVID-19.

The project demonstrates how technology can simplify healthcare operations and support digital transformation in hospitals. With further upgrades and integration in the future, the system can evolve into a complete hospital automation solution that empowers both patients and the medical staff.

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2. ResearchGate – <https://www.researchgate.net>